

Nikhil Prabhu

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EDUCATION

University of California, San Diego (UCSD)
B.S. in Computer Science | Cumulative GPA: 3.67

San Diego, CA
Expected Graduation: June 2028

SKILLS

Languages: Python, C++, TypeScript, JavaScript, Java

Robot Learning: world models, V-JEPA, PPO/MAPPO, CEM, behavioral cloning, ACT, MuJoCo

Robotics: ROS 2, SLAM, visual odometry, GPS-denied nav, YOLOX/YOLOv8, OpenCV

Systems: FastAPI, PyTorch, Electron, Next.js, Docker, WebSockets, Linux

EXPERIENCE

Part-Time Robot Operator — Verne Robotics (YC S25)

Robot-learning startup

San Francisco, CA

Jun 2026 – Jul 2026

- Operated Verne's arms collecting 300+ teleop demonstration episodes per session for policy training.
- Diagnosed transfer-time policy failures and used those traces in later world-model and imitation work.

Undergraduate Research Assistant — AI for Circuit Invention, UCSD

Research Assistant (with Prof. Prasad Gudem & Samvid AI)

San Diego, CA

March 2026 – Present

- Built an LLM + Cadence generate-simulate-evaluate loop for novel analog topologies vs SOTA benchmarks.
- Parsed Spectre logs, netlists, and metrics into structured supervision; exploring RL over the design space.

Desktop Agent Developer & Founder — UCSD Agent Development (Clue2)

Mentored by Prof. Ramamohan Paturi

San Diego, CA

Jan 2025 – Present

- Architected a cross-platform Electron + TypeScript agent that processes live meetings and executes tasks across apps.
- Engineered AssemblyAI WebSocket transcription with dual-channel capture, buffering, and fault-tolerant sessions.
- Shipped an LLM workflow engine across Gmail, Calendar, Docs, Sheets, Slack, Notion (7+); 40+ beta users, YC-startup adoption.

Lead Researcher — Foot Drop Ankle-Foot Orthosis (Georgia Tech mmWave Lab)

IEEE ICHCI 2024 Presenter & Published Author

Atlanta, GA

May 2024 – Nov 2024

- Built a 3D-printed EMG-controlled orthosis; custom actuator improved gait precision 23% ($R = 0.968$).
- First-author IEEE ICHCI 2024; refined the prototype with Georgia Tech PhD researchers.

ROBOTICS PROJECTS

DroneOS — 1st Overall, Bow Capital × DS3 × SIC

GPS-denied swarm autonomy

June 2026

- YOLOX + Norfair perception, SLAM / visual odometry, 3D Gaussian Splatting maps; MAPPO + BC swarms with no GPS or comms.

PREVUE — Pre-execution World-Model Verifier

V-JEPA + Claude planner

2026

- Imagines ~4s of pick-and-place from camera frames; injected 6 cm miss 7.3% → 94.7% after repair; 0/8 vs 6/8 across eight scenes.

EgoGoal — Unitree G1 Penalty Kick

Motion retargeting + CEM in MuJoCo

2026

- Pelvis-local retargeting + CEM: 2.1 m/s / 3.7 m dead ball → 9.94 m/s on-target center; corners within 0.01 m of crossing.

SODIUM — Autonomous Care Robot — 1st Overall, DiamondHacks

ROS 2 + Svelte 5 / Bun

April 2026

- Wake-word detection, NL routing, medication reminders, crisis alerts; in-home robot UI and real-time caregiver dashboard.

PUBLICATIONS

- **Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- **Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.